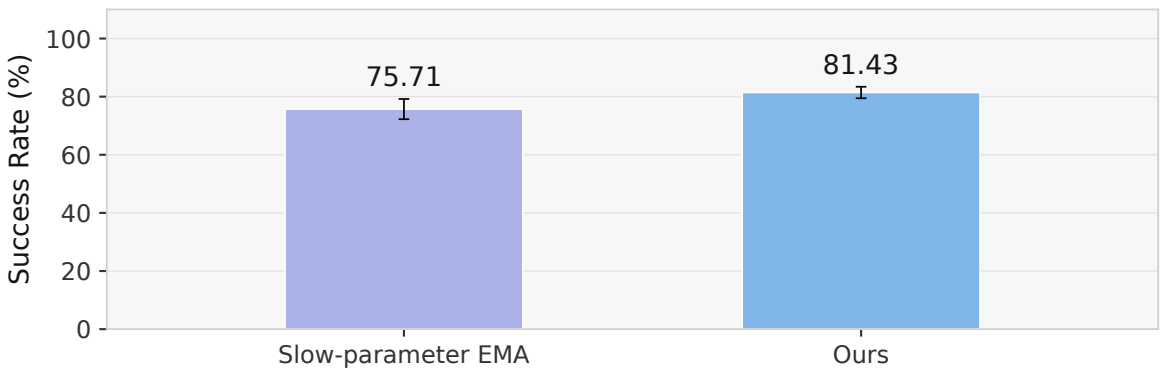
Slow-parameter EMA vs Ours

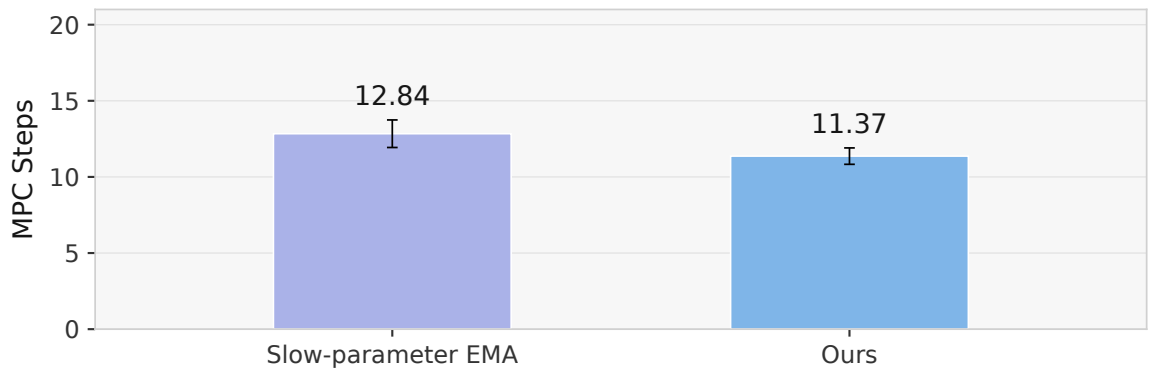
Slow-parameter EMA Ours

Seven shape domains | GD | Same original shape pretrained checkpoint

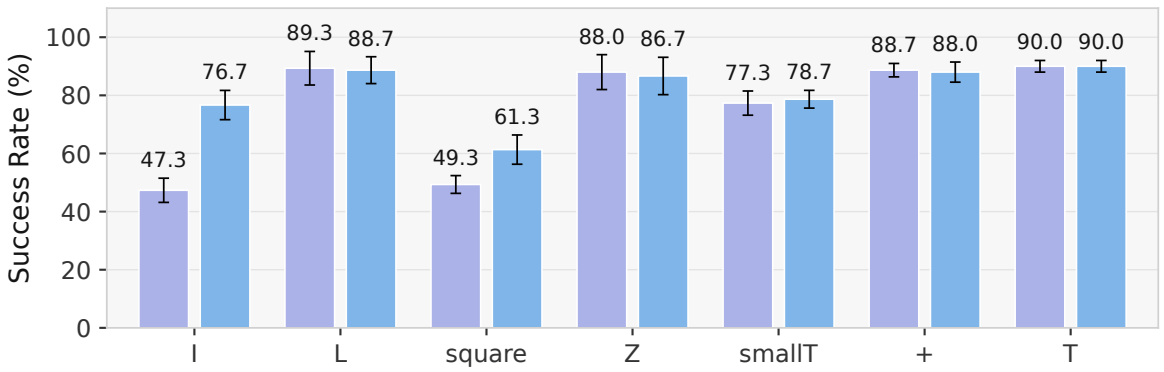
(a) Overall held-out success rate



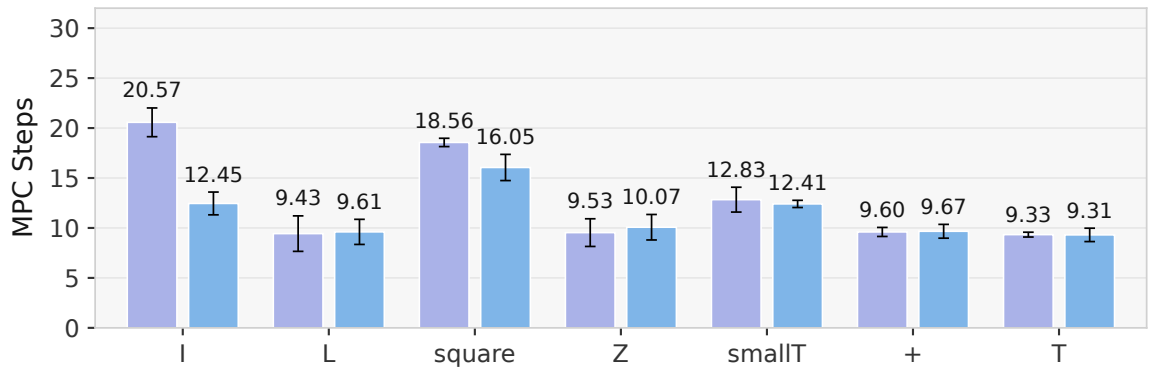
(b) Mean MPC steps (all test episodes)



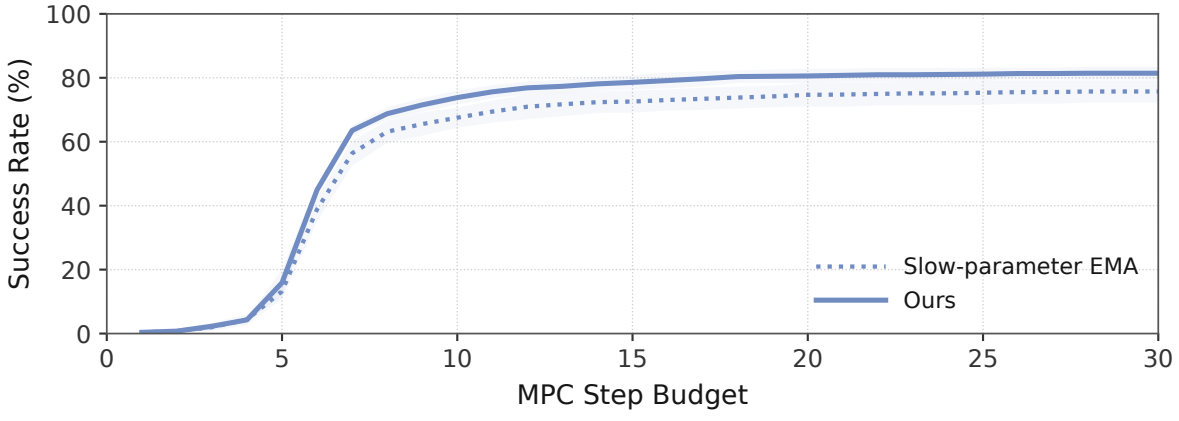
(c) Held-out success by shape



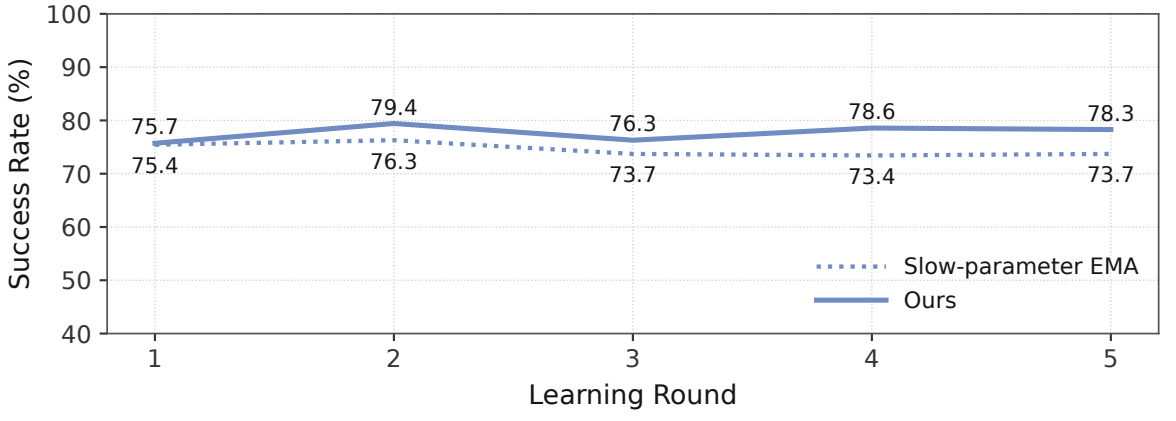
(d) Mean MPC steps by shape



(e) Held-out success within MPC budget



(f) Learning-stage success (not held-out)



Held-out tests: 1,050 episodes per method, 150 per shape; round-5 endpoints, one learning chain.

Error bars and budget shading: +/- 1 SD across three test seeds, not training repeats. Failures count as 30 MPC steps.

Panel (f): 350 online-learning episodes per round; new seed each round. It is not a fixed-test evaluation curve.